UAPOML Week 1 Assignment

1.a) Simple returns:
Month 1: (108 - 100)/100 = 8%
Month 2: (103 - 108)/108 = -4.63%
Month 3: (115 - 103)/103 = 11.65%
Month 4: (110 - 115)/115 = -4.35%

b) Log returns:
Month 1: ln(108/100) = 7.70%
Month 2: ln(103/108) = -4.74%
Month 3: ln(115/103) = 11.02%
Month 4: ln(110/115) = -4.45%

c) Comparison: The numbers match pretty closely. The approximation is least accurate for Month 3. This is because the return is the largest there. The log approximation only works good for very small returns.

2. a) Linearity: E[aX + bY] = aE[X] + bE[Y].
So, E[Rp] = (0.4 * 15%) + (0.6 * 7%) = 10.2%.

b) No, it does not change.

3. Var(X + Y) = Var(X) + Var(Y) + 2 * Cov(X, Y)
Let X = w1 * R1 and Y = w2 * R2.
Var(w1 * R1) = w1^2 * Var(R1) = w1^2 * sigma1^2.
Var(w2 * R2) = w2^2 * Var(R2) = w2^2 * sigma2^2.
Cov(w1*R1, w2*R2) = w1 * w2 * Cov(R1, R2).
So sigma_p^2 = w1^2 * sigma1^2 + w2^2 * sigma2^2 + 2 * w1 * w2 * rho12 * sigma1 * sigma2.

4. a) variance = (0.5^2 * 25^2) + (0.5^2 * 12^2) + 2(0.5)(0.5)(rho)(25)(12) = 192.25 + 150(rho).
If rho = 1: variance is 342.25, so risk is sqrt(342.25) = 18.5%.
If rho = 0: variance is 192.25, so risk is sqrt(192.25) = 13.87%.
If rho = -1: variance is 42.25, so risk is sqrt(42.25) = 6.5%.

b) It equals the weighted average only when rho = 1.
If rho = 1, variance = (w1*sig1 + w2*sig2)^2. So the square root is just w1*sig1 + w2*sig2.

c) For rho = -1, risk = |wA(25) - (1-wA)(12)| = 0
So 37(wA) = 12, wA = 12/37 = 0.324.

5. When two assets are mixed, the risk depends on correlation.
If correlation is less than 1, they do not move exactly the same way. When one goes down, the other might go up or stay flat. In the math formula, the "2 * w1 * w2 * rho * sig1 * sig2" part shrinks as rho gets smaller. This makes the total variance lower. So, their ups and downs cancel each other out a bit, making the whole portfolio safer.

6. a) Covariance Matrix: Multiply rho * sig_i * sig_j for each cell. Diagonals are just variance (sig^2).
Cell 1,1: 0.25^2 = 0.0625
Cell 2,2: 0.12^2 = 0.0144
Cell 3,3: 0.04^2 = 0.0016
Cell 1,2: 0.4 * 0.25 * 0.12 = 0.012
Cell 1,3: 0.1 * 0.25 * 0.04 = 0.001
Cell 2,3: 0.2 * 0.12 * 0.04 = 0.00096
The matrix looks like this:
[[0.0625, 0.012,   0.001]
 [0.012,   0.0144, 0.00096]
 [0.001,   0.00096, 0.0016]]

(b) Yes, it is symmetric.

7. a) X -> (14 - 4) / 22 = 0.454
Y -> (9 - 4) / 10 = 0.500
Z -> (20 - 4) / 35 = 0.457

b) Y is the best (0.50), then Z (0.457), then X (0.454). A investor wants the most return for the least risk, so they pick Y.

c) Even though Z makes the most money, it is too risky. It's better to pick Y. If the investor wants more total return, they can just borrow money and put it all into Y. That gives better returns for the same risk level compared to just holding Z.

8. a) The Global Minimum Variance portfolio is the combination of assets that has the absolute lowest possible risk. It is on the far left of the curve because you just cannot get the standard deviation any lower on the x-axis.

b) The student is wrong. The lower half of the curve is bad. For any point on the bottom half, you can find a point straight above it on the top half. That means you get more return for the exact same amount of risk.

c) The Capital Market Line is a straight line drawn from the risk-free rate on the y-axis to the highest possible tangent point. The slope of this line is the highest Sharpe ratio you can get in the whole market.

9. a) The formula is std_dev / sqrt(N). So 0.30 / sqrt(36) = 0.30 / 6 = 0.05 (or 5%). This shows our guess for the return is very noisy. If we only use historical data, our math will rely on weak guesses, leading to bad portfolios.

b) The MVO math blindly trusts the inputs. If an asset has a slightly high return due to random noise, the math gives it a massive weight. It maximizes the impact of the errors. Using machine learning helps give cleaner, smarter predictions that reduce this noise.

c) A 50-asset covariance matrix has (50 * 51) / 2 = 1275 unique numbers to estimate. This is very hard because you need a huge amount of data to guess 1275 things correctly without the matrix breaking.